

MATTHEW POOL

Software Engineer | Product Designer

Python | Kotlin | Native Android | UI/UX



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matthew-pool.github.io

PROFESSIONAL SUMMARY

Software Engineer and Product Designer (B.S. CS, 3.99 GPA) who builds accessible, scalable products from the ground up. I combine a strong technical foundation in Kotlin/Android, Python, and React/TypeScript with over a decade of customer support and Fortune 500 B2B/B2C sales experience. From architecting ML recommendation engines for mobile to designing complex OOP state management systems, I bridge the gap between technical implementation, intuitive user experiences, and overarching business goals.

EDUCATION

Bachelor of Science in Computer Science

Southern New Hampshire University | Sep 2021 – Dec 2024



Summa Cum Laude • 3.99 GPA • President's List



Certificate in Data Analysis • uCertify (2024)

Associate of Arts in General Studies

Tyler Junior College | Aug 2003 – May 2005

TECHNICAL SKILLS

Product Design

UI/UX Design, Figma, Inkscape, GIMP, Adobe Creative Suite, DaVinci Resolve, CAD, SketchUp, Responsive Design, Motion Design, UI Animation, User Flows, Information Architecture

Programming Languages

Python, JavaScript, TypeScript, Kotlin, Java, C/C++, C#, SQL

Mobile Development

Android SDK, Jetpack Compose, Material Design 3, MVVM Architecture, Coroutines, Hilt, Firebase, REST APIs

Frontend Web

React, Angular, RxJS, HTML5, CSS, SVG Animation, Vite, Bootstrap, Tailwind CSS

Backend & Databases

Node.js, Express.js, Spring Boot, MySQL, MongoDB, Mongoose (ODM), AWS DynamoDB, Firebase Firestore

Data Science & Machine Learning

Keras (TensorFlow), scikit-learn, Pandas, NumPy, Matplotlib, DNN, CNN, Reinforcement Learning (RL), Deep Q-Networks (DQN)

Gaming & Graphics

Unity, OpenGL/GLSL, 2D/3D Graphics, Story-telling, Game Design & Development

Testing & QA

Postman, pytest, JUnit, MockK, Mockito, Espresso, Robolectric, Jacoco, Jest, Selenium

Infrastructure & Security

Docker/Docker Compose, AWS, CI/CD, Git/GitHub Actions, Markdown, Documentation, JWT, OWASP, Snyk, Gradle, Maven

TECHNICAL PROJECTS

Flick (Android)	<i>Kotlin, Jetpack Compose, MVVM, Hilt, Firebase, GitHub Actions</i>	Leveraged the Archive.org API to build a highly scalable, zero-infrastructure-cost video library. Paired this with a proprietary ML recommendation engine and asynchronous caching/preloading to deliver a low-latency, real-time streaming experience.
Wormhole	<i>Python, OOP, State Management</i>	Architected a complex interactive Python terminal game utilizing advanced object-oriented programming, dynamic state management, and robust game loop logic.
Full-Stack Travel Platform	<i>Angular, TypeScript, Docker, AWS DynamoDB</i>	Engineered an Angular SPA with a Dockerized backend, orchestrating a full DB migration from MongoDB to AWS.
The Existence Paradox	<i>React, TypeScript, Tailwind CSS</i>	Interactive web application featuring dynamic state management and a complex decision-tree navigation system.
Financial Security Audit	<i>Java, Spring Boot, OWASP, AES-256</i>	Refactored a legacy REST API to implement encryption at rest, mitigating injection vulnerabilities to achieve OWASP compliance.
Machine Learning Models	<i>Python, Keras (TensorFlow), scikit-learn, NumPy</i>	Trained and tuned a CNN for image classification and a Deep Q-Network (DQN) for reinforcement learning pathfinding.
Product Design & UX	<i>Figma, Wireframes, Prototyping</i>	End-to-end UX design for mobile and web apps: personas, wireframes, hi-fidelity interactive prototypes.
Professional Portfolio	<i>HTML5, CSS3, Vanilla JS</i>	Designed and hand-coded a zero-dependency responsive portfolio featuring a custom dark mode and complex SVG animations.

More projects and details at matthew-pool.github.io

PROFESSIONAL EXPERIENCE

Lead Software Engineer & Product Designer

Jan 2025 – Present

App.Buddy (Independent Venture)

- **Mobile Product Strategy:** Architected and launched **Flick**, a native Android video streaming platform — designed in Figma, built in Kotlin (MVVM), and deployed to Google Play with a proprietary ML recommendation engine.
- **Backend & Logic Engineering:** Architected **Wormhole**, a Python-based application, utilizing advanced object-oriented programming (OOP), state management, and complex logic design.
- **Web & Frontend Architecture:** Engineered a zero-dependency personal portfolio platform utilizing pure HTML/CSS/JS, and built **Existence Paradox**, a React/TypeScript application featuring dynamic state management and complex decision-tree routing.
- **Development Velocity & Prototyping:** Leveraged generative AI toolchains to accelerate end-to-end development — rapidly iterating on high-fidelity UI assets, generating boilerplate code, and scaling solo output across all applications.
- **CI/CD & QA:** Built robust CI/CD pipelines via GitHub Actions automating unit (JUnit/MockK) and UI tests (Espresso) to ensure stable production deployments.

Outbound Sales & Account Management (2004–2009)

★ Recurring Top Performer

Sales Representative • T.C.I.M. Services (B2B/B2C Outbound Sales)

- Conducted high-volume outbound sales calls and account management on behalf of Fortune 500 clients including American Express, AT&T, QVC, Citibank, Blue Cross Blue Shield, among others.
- Consistently ranked among the top performers in a rolling pool of 100+ representatives, exceeding KPIs across acquisition, upsell, and retention metrics.
- Developed call scripts adopted across the team and actively trained junior sales representatives on objection handling and closing techniques.

Technical Support, IT Operations & Inbound Sales (2011–2013)

Technical Support Specialist & Desktop Support • Sutherland | New Electronics Warranty | TeleNetwork

- Diagnosed and resolved complex hardware, software, VoIP, and ISP issues via remote desktop tools; documented resolutions in CRM knowledge bases to improve team efficiency.
- Translated complex technical issues into clear guidance for diverse consumer audiences.
- Identified upsell and upgrade opportunities during support interactions, consistently meeting or exceeding inbound sales and satisfaction targets.
- Clients included Frontier Communications and DirecTV premium support customers.

Shift Management & Customer Service (1997–2003 / 2009–2010)

 Management & Ops

Shift Manager & Retail Associate • Kmart / Fast Food & Customer Service (East TX)

- Directed workflow, delegated tasks, and trained new hires across fast-paced retail and food service environments.
- Regularly trusted to close stores independently at night, maintaining full financial accountability for cash registers and daily deposits.
- Managed inventory, restocking, and merchandising to optimize sales floor presentation.
- Operated cash registers and promoted store membership programs at store entrance.

Independent Projects & Family Leave (2014–2021)

Construction & Primary Caregiver

- Developed woodworking skills while working at Diamond G Cedar (family-owned business) and doing independent research online.
- Applied project management skills to successfully design and construct a 3-bedroom residential home from an existing woodshed, minimizing project costs under strict budgetary and labor constraints by carefully salvaging and cleaning raw materials — one nail at a time.
- Served as caregiver for a family of four and helped facilitate spouse's successful return to college to receive her BBA in Finance, prior to returning to college myself to earn my Computer Science degree.